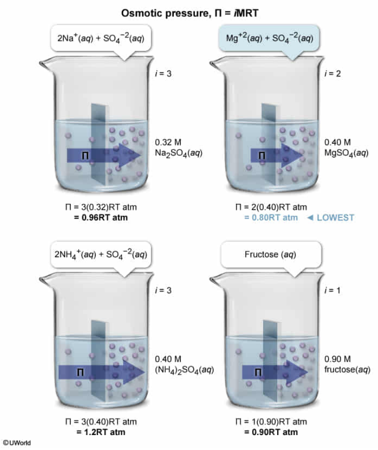
If each of the following is an ideal solution at 37 °C, the lowest osmotic pressure, equaling 20 atm, is produced by:

- A. 0.32 M Na_2SO_4. [16%]
- ✔ B. 0.40 M $MgSO_4$. [52%]
- C. 0.40 M $(NH_4)_2SO_4$. [12%]
- D. 0.90 M fructose. [18%]

Correct
52% answered correctly

Explanation:



When two solutions with different solute concentrations are separated by a **semi-permeable membrane**, the solvent from the solution of lower solute concentration will diffuse across the membrane into the solution of higher solute concentration until the solution **concentrations equalize**. This behavior is called osmosis. The **diffusion of the solvent** during **osmosis** produces an **osmotic pressure** against the membrane, which is calculated for an ideal solution by the relationship:

$$\Pi = iMRT$$

where Π is the osmotic pressure, i is the van 't Hoff factor (the number of species a molecule will form when dissolved in solution), M is the solute molar concentration (mol/L), R is the ideal gas constant (0.0821 L·atm/mol·K), and T is the solution temperature (Kelvin).

When the ionic $MgSO_4$ is dissolved in water, it dissociates into two ions (Mg^{+2} and SO_4^{-2}), making the van 't Hoff factor equal to 2. A solution that has 0.40 M $MgSO_4$, therefore, produces a solution that has 0.40 M Mg^{+2} and 0.40 M SO_4^{-2} and gives a solution with an osmotic pressure expressed generally by:

$$\Pi$$

At 37 °C (310 K), RT = 25 L·atm/mol, yielding a pressure of 20 atm.

$$\Pi = 0.80$$

The other solutions involve compounds for which the product of the van 't Hoff factor and the molar solute concentration produce a higher osmotic pressure; therefore, 0.40 M $MgSO_4$ exhibits the lowest osmotic pressure.

(Choice A) Aqueous Na_2SO_4 dissociates into 2 Na^+ and SO_4^{-2} (i = 3). Therefore, $\Pi = 3(0.32)RT = 0.96RT$ atm.

(Choice C) Aqueous $(NH_4)_2SO_4$ dissociates into 2 NH_4^+ and SO_4^{-2} (i = 3). As such, $\Pi = 3(0.40)RT = 1.2RT$ atm.

(Choice D) Fructose dissolved in water does not dissociate (i = 1). Consequently, $\Pi = 1(0.90)RT = 0.90RT$ atm.

Educational objective:
Osmosis is the diffusion of solvent across semi-permeable membrane from a solution of lower solute concentration into a solution of higher solute concentration until the solution concentrations equalize. During osmosis, solvent diffusion imparts an osmotic pressure that is calculated by $\Pi = iMRT$, which is the product of the van 't Hoff factor, the solute molar concentration, the ideal gas constant, and the absolute temperature.

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